

Institute/Department	UNIVERSITY INSTITUTE OF ENGINEERING (UIE)	Program	Bachelor of Engineering (Hons.) (Computer Science and Engineering) (Cyber Security) (CS230)
Master Subject Coordinator Name:	Arulkumar S	Master Subject Coordinator E-Code:	E18999
Course Name	Database Management System	Course Code	25CSH-236

Lecture	Tutorial	Practical	Self Study	Skilling	TC	TGT	TGP	Studio	Credit	Subject Type
3	0	2	0	2	0	0	0	0	4.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Hybrid	Hybrid (HYB)

Mission of the Department	M1 : To provide relevant, rigorous and contemporary curriculum and aligned assessment system to ensure effective learning outcomes for engineering technologies. M2: To provide platform for industry engagement aimed at providing hands-on training on advanced technological and business skills to our students. M3: To provide opportunities for collaborative, interdisciplinary and cutting-edge research aimed at developing solutions to real life problems M4: To imbibe quest for innovation, continuous learning and zeal to pursue excellence through hard work and problem-solving approach M5: To foster skills of leadership, management, communication, team spirit and strong professional ethics in all academic and societal endeavors of our students
Vision of the Department	To be recognized as a centre of excellence for Computer Science & Engineering education and research, through effective teaching practices, hands-on training on cutting edge computing technologies and excellence in innovation, for creating globally aware competent professionals with strong work ethics whom would be proficient in implementing modern technology solutions and shall have entrepreneurial zeal to solve problems of organizations and society at large.

Program Educational Objectives(PEOs)	
PEO1	To be able to explore areas of research, technology application & innovation and make a positive impact in different types of institutional settings such as corporate entities, government bodies, NGOs, inter-government organizations, & start-ups.
PEO2	To be able to design, and implement technology and computing solutions to organizational problems, effectively deploy knowledge of engineering principles, demonstrate critical thinking skills & make the intellectual connections between quantitative and qualitative tools, theories, and context to solve the organizational problems
PEO3	To be able to work with, lead & engage big and small teams comprising diverse people in terms of gender, nationality, region, language, culture & beliefs. To understand stated and unstated differences of views, beliefs & customs in diverse & interdisciplinary team settings
PEO4	To be able to continuously learn and update one's knowledge, engage in lifelong learning habits and acquire latest knowledge to perform in current work settings
PEO5	To continuously strive for justice, ethics, equality, honesty, and integrity both in personal and professional pursuits. Able to understand and conduct in a way that is responsible and respectful.

Program Specific Outcomes(PSOs)	
PSO1	The graduate student shall be able to analyze and evaluate systems with respect to maintaining operations in the presence of risks and threats, and also communicate the human role in security systems with an emphasis on ethics, social engineering vulnerabilities and training.
PSO2	The graduate student shall be able to conduct an Cyber Security risk assessment using Cryptographic measures and be able to design and develop operational and strategic security policies.
PSO3	The graduate student shall be able to assess the Ethical Ramification of working in Cyber Security, information assurance, and cyber/computer forensics software/tools and architecture

Program OutComes(POs)

PO1	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, societal, and environmental considerations.
PO4	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of the information to provide valid conclusions.
PO5	Create, select, and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as member and leader in a team, to manage projects and in multidisciplinary environments
PO12	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context to technological change.
PO13	Demonstrate the capability to apply analytic thought to a body of knowledge, including the analysis and evaluation of policies, and practices. Identify relevant assumptions or implications, logical flaws and loopholes in the presented arguments.
PO14	Demonstrate to create, perform, or think in different and diverse ways about the given scenario. Innovate and perform tasks in a better manner, view a problem or a situation from multiple perspectives, think 'out of the box' and generate solutions to complex problems in unfamiliar contexts
PO15	Demonstrate the ability to identify with or understand the perspective, experiences, or points of view of another individual or group, and to identify and understand other people's emotions
PO16	Demonstrate the ability to participate in community-engaged services/ activities for promoting the well-being of society
PO17	Demonstrate the acquisition of knowledge of the values and beliefs of multiple cultures, capability to effectively engage in a multicultural group/society and interact respectfully with diverse groups and gender sensitivity and adopting a gender-neutral approach, as also empathy for the less advantaged and the differently-abled including those with learning disabilities.

Text Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Database Management Systems	Carlos Coronel, Steven Morris	13th Edition	Cengage Learning	2018
2	Database System Concepts	Abraham Silberschatz, Henry F. Korth, S. Sudarshan	7th Edition	McGraw-Hill Education	2019
3	Fundamentals of Database Systems	Ramez Elmasri, Shamkant B. Navathe	7th Edition	Pearson Education / Addison Wesley	2015

Reference Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	An Introduction to Database Systems	C. J. Date	8th Edition	Addison Wesley	2003
2	Database Principles: Fundamentals of Design, Implementation, and Management	Carlos Coronel, Steven Morris, Peter Rob	9th Edition	Cengage Learning	2014
3	Database Systems: A Practical Approach to Design, Implementation and Management	Thomas M. Connolly, Carolyn E. Begg	5th Edition	Addison-Wesley	2010

Course OutCome	
SrNo	OutCome
CO1	Understand the database concept, system architecture and role of Database Administrator
CO2	Design a database for an organization using data model and SQL queries to ensure for an organization for efficient management and retrieval.
CO3	Apply Relational Algebra and Relational Calculus Queries to the database of an organization. esign a database for an organization using SQL queries to ensure
CO4	Implement packages, procedures, and triggers to enhance functionality, automate processes, and maintain data integrity in the organization's database using PL/SQL.
CO5	Implement transaction processing and concurrency ontrl to ensure data consistency, integrity, and fficient multi-user access in a database system

Lecture Plan Preview-Theory						
Unit No	SessionNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	CO - BT Level Mapping
1	1	Chapter 1.1 - Overview of Databases	Database concepts, DBMS definition, comparison with traditional file systems	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO1 - BT1
1	2	Chapter 1.1 - Overview of Databases	Three-level ANSI-SPARC Architecture; Advantages and Disadvantages of DBMS	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO1 - BT2
1	3	Chapter 1.1 - Overview of Databases	Data Independence (Logical & Physical); DBA and Responsibilities of DBA	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO1 - BT2
1	4	Chapter 1.1 - Overview of Databases	Relational Data Structure: Keys, Relations, Attributes, Schema and Instances	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO1 - BT2
1	5	Chapter 1.1 - Overview of Databases	Referential Integrity, Entity Integrity; Introduction to importance of Data Models	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO1 - BT2
1	6	Chapter 1.2 - Data Models and SQL	Basic building blocks, Business rules, Evolution of data models, Degrees of data abstraction; Relational, Network, Hierarchical Models	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO2 - BT2
1	7	Chapter 1.2 - Data Models and SQL	ER Model: Design, issues, mapping constraints, ER Diagram; Comparison of Models; Introduction to UML	,T-Database System Concepts,T-Fundamentals of Database Syste,R-Database Systems: A Practical	PPT,Simulati on,Video Lecture	CO2 - BT3
1	8	Chapter 1.2 - Data Models and SQL	SQL: Data Definition, datatypes, schema definition, creating tables	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO2 - BT3
1	9	Chapter 1.2 - Data Models and SQL	SQL: Basic structure of queries, DML operations – INSERT, UPDATE, DELETE	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT,Simulati on,Video Lecture	CO2 - BT3
1	10	Chapter 1.2 - Data Models and SQL	DDL commands (CREATE, ALTER, DROP), Set Operations, Aggregate Functions, NULL values	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT	CO2 - BT3
1	11	Chapter 1.3 - Relational Languages	Relational Algebra: Introduction, Syntax, Semantics, Selection and Projection	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT	CO3 - BT2
1	12	Chapter 1.3 - Relational Languages	Relational Algebra: Additional operators, Grouping, Ungrouping, Relational Comparisons	,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Sy	PPT	CO3 - BT3

1	13	Chapter 1.3 - Relational Languages	Relational Calculus: Tuple Calculus (TRC), Domain Calculus (DRC)	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO3 - BT3
1	14	Chapter 1.3 - Relational Languages	Calculus vs Algebra, Computational capabilities; Practice problems on RA and RC	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO3 - BT4
1	15	Chapter 1.3 - Relational Languages	Unit 1 Revision: Key concepts recap, MST-I preparation, Q&A session	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT, Simulation	CO1 - BT2; CO2 - BT2; CO3 - BT2
2	16	Chapter 2.1 - Functional Dependencies and Normalization	Functional Dependencies: definition, types, Armstrong's axioms	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO2 - BT2
2	17	Chapter 2.1 - Functional Dependencies and Normalization	Full Functional Dependency (FFD), Transitive Dependency (TD), Decomposition	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO2 - BT2
2	18	Chapter 2.1 - Functional Dependencies and Normalization	Join Dependency (JD), Multi-Valued Dependency (MVD)	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO2 - BT2
2	19	Chapter 2.1 - Functional Dependencies and Normalization	Normal Forms: 1NF and 2NF – definitions and examples	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Systems: A Practical	PPT	CO2 - BT3
2	20	Chapter 2.1 - Functional Dependencies and Normalization	Normal Forms: 3NF and BCNF – definitions and examples	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Systems: A Practical	PPT	CO2 - BT3
2	21	Chapter 2.1 - Functional Dependencies and Normalization	Normal Forms: 4NF and 5NF; Denormalization: concepts and use cases	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Systems: A Practical	PPT	CO2 - BT4
2	22	Chapter 2.2 - Database Security	Database Security: Introduction, Threats – SQL Injection, Unauthorized Access	,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO1 - BT2
2	23	Chapter 2.2 - Database Security	Security Counter Measures: Authentication, Authorization, Encryption, Auditing	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO1 - BT3
2	24	Chapter 2.3 - Control Structures	PL/SQL Control Structures: Conditional control – IF-THEN-ELSE, Sequential control	,T-Database Management Systems,R-Database Principles: Fundament	PPT	CO4 - BT3
2	25	Chapter 2.3 - Control Structures	Iterative control: LOOP, WHILE, FOR loops; Nested control structures	,T-Database Management Systems,R-Database Principles: Fundament	PPT	CO4 - BT3
2	26	Chapter 2.3 - Control Structures	Cursors: Implicit and Explicit cursors, Cursor attributes, Cursor FOR loops	,T-Database Management Systems,R-Database Principles: Fundament	PPT	CO4 - BT3
2	27	Chapter 2.3 - Control Structures	Views: Creating and managing views; Comparison between Tables and Views	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Systems	PPT	CO4 - BT3
2	28	Chapter 2.3 - Control Structures	Advanced SQL: Joins – INNER, LEFT, RIGHT, FULL OUTER, NATURAL; Indexes	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Systems	PPT	CO2 - BT3

2	29	Chapter 2.3 - Control Structures	SQL Subqueries and Nested Queries: ANY, ALL, IN, EXISTS, UNION, INTERSECT	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT	CO3 - BT4
2	30	Chapter 2.3 - Control Structures	Unit 2 Revision: Normalization, Security, PL/SQL Controls; MST-II preparation	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO2 - BT2
3	31	Chapter 3.1 - Packages, Procedures and Triggers	Stored Procedures: Parts of procedures, Parameter modes (IN, OUT, IN OUT), Advantages	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO5 - BT2
3	32	Chapter 3.1 - Packages, Procedures and Triggers	Creating and executing procedures; PL/SQL Functions vs Procedures	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO5 - BT3
3	33	Chapter 3.1 - Packages, Procedures and Triggers	Triggers: Syntax, types – BEFORE/AFTER, ROW/STATEMENT, INSERT/UPDATE/DELETE triggers	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO5 - BT3
3	34	Chapter 3.1 - Packages, Procedures and Triggers	Packages: Bodiless packages, Package specification and body, Advantages of packages	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO5 - BT3
3	35	Chapter 3.1 - Packages, Procedures and Triggers	Aggregate functions, NULL values handling, Nested subqueries in PL/SQL	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT	CO5 - BT3
3	36	Chapter 3.2 - Transaction Management and Concurrency Control	Introduction to Transaction Processing; ACID properties – Atomicity, Consistency, Isolation, Durability	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT2
3	37	Chapter 3.2 - Transaction Management and Concurrency Control	Serializability: Conflict serializability, View serializability; Recoverability	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT2
3	38	Chapter 3.2 - Transaction Management and Concurrency Control	Need for Concurrency Control; Locking techniques – Shared/Exclusive locks, Two-Phase Locking (2PL)	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT3
3	39	Chapter 3.2 - Transaction Management and Concurrency Control	Timestamp-based methods, Optimistic concurrency control; Granularity of data items	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT3
3	40	Chapter 3.2 - Transaction Management and Concurrency Control	Deadlock handling: Detection, Prevention, Avoidance; Weak levels of consistency; Insert and Delete operations	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT4
3	41	Chapter 3.3 - Database Recovery	Introduction to Database Recovery; Types of errors; Need for recovery	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT2
3	42	Chapter 3.3 - Database Recovery	Log-based recovery; Recovery and atomicity; Storage structure for recovery	,T-Database System Concepts,T-Fundamentals of Database System,R-An Introduction to Database Sy	PPT	CO4 - BT3

3	43	Chapter 3.3 - Database Recovery	Recovery with concurrent transactions; Buffer management; Failure with loss of non-volatile storage	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO4 - BT3
3	44	Chapter 3.3 - Database Recovery	Validation-based protocols; Insert and Delete operations under concurrency control	,T-Database System Concepts,T-Fundamentals of Database System, R-An Introduction to Database Systems	PPT	CO4 - BT4
3	45	Chapter 3.3 - Database Recovery	Comprehensive Revision: All three units; ETE preparation; Case study discussion	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Systems: A Practical	PPT	CO1 - BT5; CO4 - BT5; CO5 - BT5

Lecture Plan Preview-Practical

Unit No	SessionNo	Experiment Name	Text/ Reference Books	Pedagogical Tool**	CO - BT Level Mapping
1	1	Installation of MySQL and practicing basic SQL commands – SELECT, FROM, WHERE, ORDER BY	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT	CO1 - BT1
1	2	Create Tables and Implement DDL Queries – CREATE, ALTER, DROP, TRUNCATE, RENAME	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT	CO2 - BT3
1	3	Implement DML Queries – INSERT, UPDATE, DELETE, SELECT with conditions	,T-Database Management Systems,T-Database System Concepts,R-Database Principles: Fundament	PPT,Simulation	CO2 - BT3
1	4	Enforce Integrity Constraints – Domain, Primary Key, Foreign Key, NOT NULL, UNIQUE, DEFAULT, CHECK	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	Video Lecture	CO3 - BT3
1	5	Practice session – Combined DDL + DML + Constraints on a mini database (e.g. Student / Employee table)	,T-Database Management Systems,T-Database System Concepts,R-An Introduction to Database Sy	PPT	CO2 - BT3; CO3 - BT4
2	6	Implement Queries on Joins – Natural Join, Left Outer, Right Outer, Full Outer Join	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO4 - BT3
2	7	Implement Procedures and Cursors – Stored procedures with IN/OUT parameters, explicit cursors	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO4 - BT3
2	8	Implement Queries using Aggregate Functions – COUNT, SUM, AVG, MAX, MIN with GROUP BY, HAVING, ORDER BY	,T-Database Management Systems,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO3 - BT3
2	9	Implement Queries using ANY, ALL, IN, EXISTS, UNION, INTERSECT with nested subqueries	,T-Database System Concepts,R-Database Systems: A Practical	PPT	CO3 - BT4
2	10	Practice session – Combined Joins + Subqueries + Aggregate Functions on a real-world schema (e.g. Library / Company DB)	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO3 - BT4; CO4 - BT4
3	11	Implement Triggers – INSERT trigger, DELETE trigger, UPDATE trigger with BEFORE/AFTER types	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Principles: Fundament	PPT	CO4 - BT3; CO5 - BT3
3	12	Design Case Study – Company Database / Student Progress Monitoring System / Book Publishing Company	,T-Database System Concepts,T-Fundamentals of Database System, R-Database Systems: A Practical	PPT	CO5 - BT4

3	13	Mini Project Part 1 – ER Design and Table Creation for chosen project (Library / Retail / Railway DB)	,T-Database System Concepts,T-Fundamentals of Database Syste,R-Database Principles: Fundament	PPT	CO5 - BT4
3	14	Mini Project Part 2 – Implement Queries, Constraints, Joins and Triggers on the project database	,T-Database System Concepts,T-Fundamentals of Database Syste,R-Database Systems: A Practical	PPT	CO5 - BT6
3	15	Mini Project Part 3 – Final demonstration, Viva and complete documentation of the project	,T-Fundamentals of Database Syste,R-Database Systems: A Practical	PPT	CO5 - BT5

Assessment Model			
Sr No	Exam Name	Max Marks	Weighted Marks
1	Attendance Marks	2	2
2	MST-1 Hybrid	20	5
3	MST-2 Hybrid	20	5
4	Practical assessment-1	30	7.5
5	Practical assessment-2	30	7.5
6	Practical assessment-3/Project Evaluation	30	7.5
7	Case Study or Course Project	20	5
8	Practical Mid Term	30	7.5
9	Surprise Test or Quiz	12	3
10	Assignment Mark	12	5
11	End Term Theory Exam	60	25
12	Practical End Term	40	20